

## DARWIN AND EVOLUTION

# Module map

- Module 12: Darwin and Evolution
- Connected lab: lab-13\_darwin-evolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 12

Darwin and Evolution

1

Historical  
explanation  
for  
adaptation

2

Variation  
and  
heritability

3

Natural  
selection  
mechanism

4

Evidence  
for  
common  
ancestry

5

Fitness as  
reproductive  
success

### Lab evidence

lab-13\_darwin-evolution.md

## DARWIN AND EVOLUTION

# Learning objectives

- Explain natural selection using variation, heritability, and reproductive success.
- Define fitness in evolutionary terms.
- Distinguish individual change from population evolution.
- Use multiple evidence types to support common ancestry.
- Apply natural selection to a concrete trait example.

**OBJECTIVE 1**

Explain natural selection using variation, heritability, and reproductive success.

**OBJECTIVE 2**

Define fitness in evolutionary terms.

**OBJECTIVE 3**

Distinguish individual change from population evolution.

**OBJECTIVE 4**

Use multiple evidence types to support common ancestry.

**OBJECTIVE 5**

Apply natural selection to a concrete trait example.

## DARWIN AND EVOLUTION

# Topic sequence

- Historical explanation for adaptation: Darwinian evolution explains adaptation through descent with modification.
- Variation and heritability: Natural selection requires variation, heritability, and unequal reproductive success.
- Natural selection mechanism: Fitness means relative reproductive success in a particular environment.
- Evidence for common ancestry: Fossils, homology, biogeography, and molecular evidence support common ancestry.
- Fitness as reproductive success: Evolution changes populations over generations, not individual organisms during life.

01

**Historical explanation for adaptation**

Darwinian evolution explains adaptation through descent with modification.

02

**Variation and heritability**

Natural selection requires variation, heritability, and unequal reproductive success.

03

**Natural selection mechanism**

Fitness means relative reproductive success in a particular environment.

04

**Evidence for common ancestry**

Fossils, homology, biogeography, and molecular evidence support common ancestry.

05

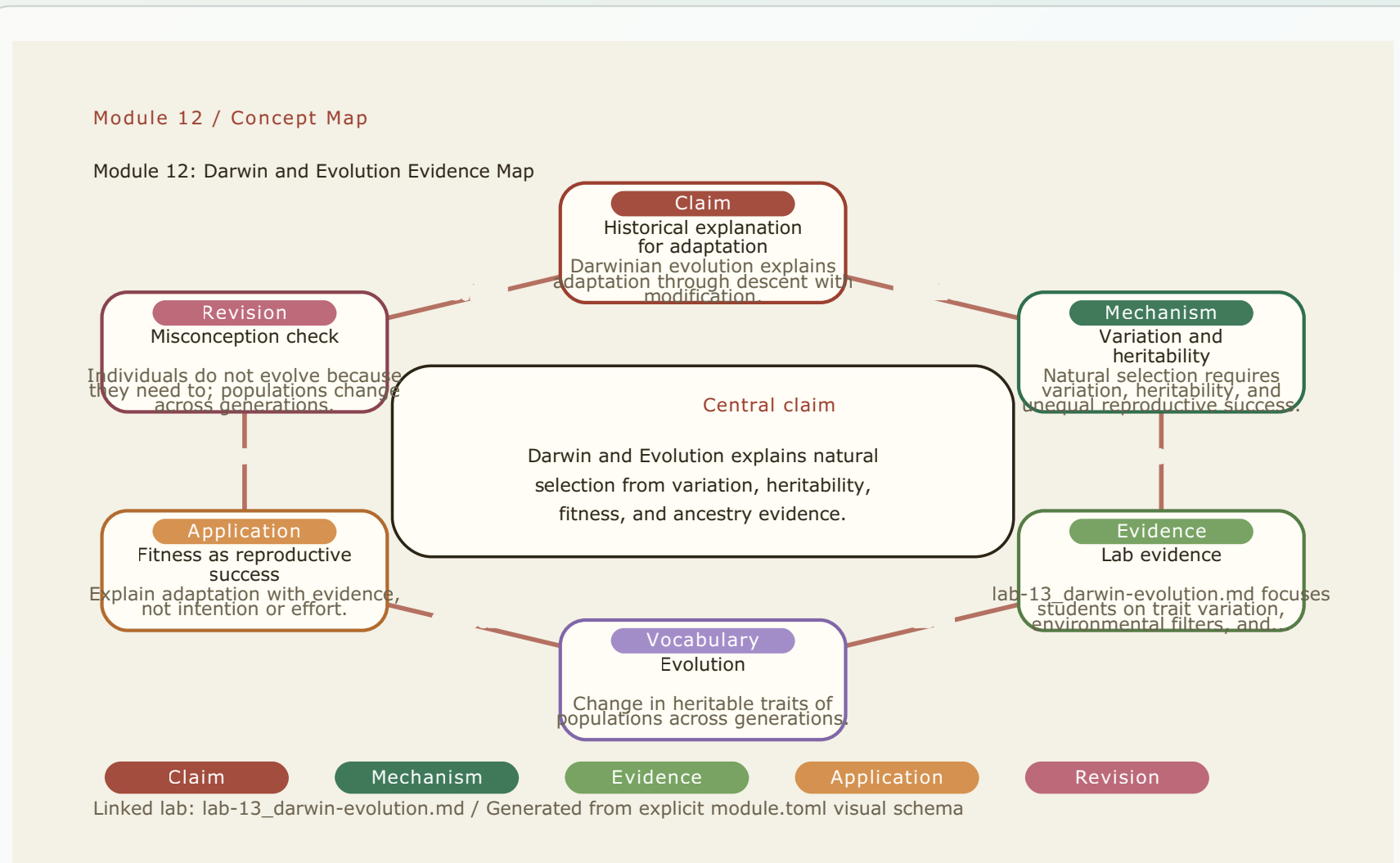
**Fitness as reproductive success**

Evolution changes populations over generations, not individual organisms during life.

## DARWIN AND EVOLUTION

# Module 12: Darwin and Evolution Evidence Map

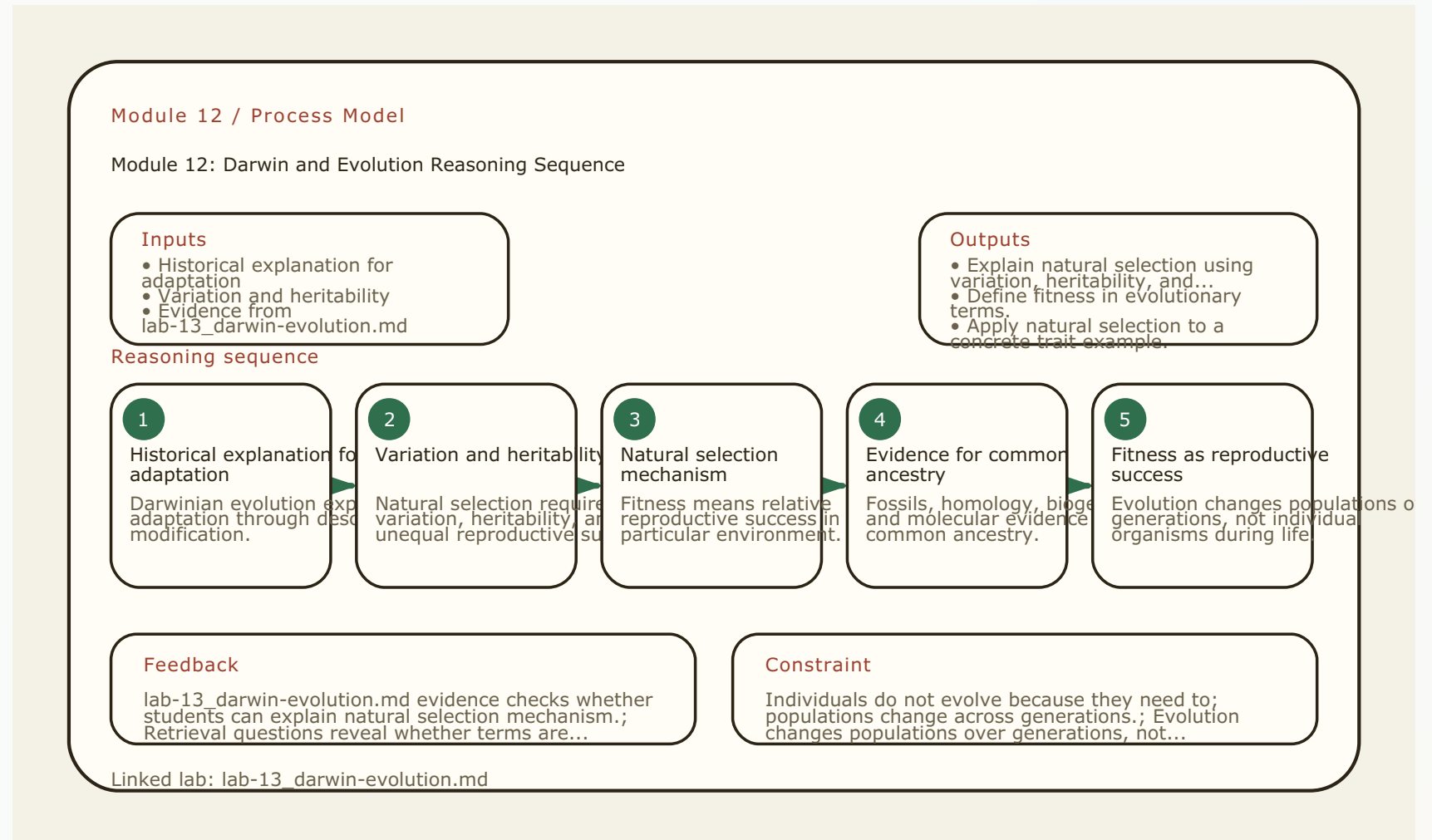
- Module focus: Darwin and Evolution
- Central claim: Darwin and Evolution explains natural selection from variation, heritability, fitness, and ancestry evidence.
- Key nodes to connect: Historical explanation for adaptation, Variation and heritability, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-13\_darwin-evolution.md supplies an evidence surface for the map.



## DARWIN AND EVOLUTION

# Module 12: Darwin and Evolution Reasoning Sequence

- Module focus: Darwin and Evolution
- Trace the model: Historical explanation for adaptation -> Variation and heritability -> Natural selection mechanism -> Evidence for common ancestry
- Inputs become outputs: Historical explanation for adaptation, Variation and heritability -> Explain natural selection using variation, heritability, and reproductive success., Define fitness in evolutionary terms.
- Feedback check: lab-13\_darwin-evolution.md evidence checks whether students can explain natural selection mechanism.
- Lab connection: lab-13\_darwin-evolution.md gives students a process to observe or test.



## DARWIN AND EVOLUTION

# Terms as evidence handles

- Evolution: Change in heritable traits of populations across generations.
- Natural selection: Nonrandom differences in survival or reproduction tied to heritable traits.
- Adaptation: Heritable trait shaped by selection in a context.
- Fitness: Relative reproductive success.
- Common ancestry: Shared origin of lineages through descent.
- Homology: Similarity due to shared ancestry.

**EVOLUTION**

Change in heritable traits of populations across generations.

**NATURAL SELECTION**

Nonrandom differences in survival or reproduction tied to heritable traits.

**ADAPTATION**

Heritable trait shaped by selection in a context.

**FITNESS**

Relative reproductive success.

**COMMON ANCESTRY**

Shared origin of lineages through descent.

**HOMOLOGY**

Similarity due to shared ancestry.

## DARWIN AND EVOLUTION

# Lab connection

- Lab file: lab-13\_darwin-evolution.md
- Lab evidence should test or illustrate:  
Natural selection mechanism
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**

Historical explanation  
for adaptation

**EVIDENCE**

lab-13\_darwin-  
evolution.md

**CLAIM**

Explain natural  
selection using  
variation, heritability,  
and reproductive  
success.

## DARWIN AND EVOLUTION

# Contrast check

- Do not stop at: Darwinian evolution explains adaptation through descent with modification.
- Stronger explanation: Evolution changes populations over generations, not individual organisms during life.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Darwinian evolution explains adaptation through descent with modification.

**DEEPER**

Evolution changes populations over generations, not individual organisms during life.

## DARWIN AND EVOLUTION

# Module 12: Darwin and Evolution Retrieval and Lab Check

- Module focus: Darwin and Evolution
- Retrieval prompt: What are the required ingredients for natural selection?
- Answer check: Explain natural selection using variation, heritability, and reproductive success.
- Required terms: Evolution, Natural selection, Adaptation
- Lab connection: lab-13\_darwin-evolution.md; lab-13\_darwin-evolution.md supplies evidence for trait variation, environmental filters, and selection evidence.

## Module 12 / Retrieval Card

Module 12: Darwin and Evolution

Retrieval and Lab Check

## Routine

Cover notes -> answer aloud  
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

What are the required ingredients for natural selection?

Check: Explain natural selection using variation, heritability, and reproductive success.

Why does natural selection act on existing variation?

Check: Define fitness in evolutionary terms.

What does evolutionary fitness mean?

Check: Distinguish individual change from population evolution.

Why do populations evolve rather than individuals?

Check: Use multiple evidence types to support common ancestry.

## Terms to use

Evolution, Natural selection, Adaptation, Fitness, Common ancestry

## Lab connection

lab-13\_darwin-evolution.md supplies evidence for trait variation, environmental filters, and selection evidence.

Intro biology routine: claim, evidence, reasoning, revision.

## DARWIN AND EVOLUTION

# Practice quiz bridge

- 1. Natural selection requires all of the following EXCEPT:
- 2. In evolutionary terms, an organism's "fitness" means its:
- 3. The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:
- 4. "An individual giraffe stretched its neck during its life and so it evolved a longer neck." Why is this wrong?

**QUESTION 1****Answer first**

Natural selection requires all of the following EXCEPT:

**QUESTION 2****Answer first**

In evolutionary terms, an organism's "fitness" means its:

**QUESTION 3****Answer first**

The forelimb bones of humans, whales, and bats share the same underlying arrangement. This is evidence of:

**QUESTION 4****Answer first**

"An individual giraffe stretched its neck during its life and so it evolved a longer neck." Why is this wrong?

## DARWIN AND EVOLUTION

# Synthesis and exit ticket

- Exit claim: explain historical explanation for adaptation using evidence from lab-13\_darwin-evolution.md.
- Must include: Evolution, Natural selection, Adaptation.
- Revision prompt: what evidence would change your explanation?

**CLAIM**

Historical explanation for adaptation

**EVIDENCE**

lab-13\_darwin-evolution.md

**REVISION**

What would change your mind?